501(C)(3) NON-PROFIT · OPEN SOURCE · AUSTIN, TEXAS

Bitcoin has no annual report. So we publish one.

Bitcoin Data Labs measures Bitcoin in public. We turn its open record (every commit and code review, every Lightning channel, every block) into datasets, dashboards, reports and research papers that anyone can check, on a fixed schedule and with no paywall.

2	25+	10+	17 yrs	0
research papers on arXiv, October 2026	automated jobs publishing daily, weekly and monthly	open datasets in public repositories	of Bitcoin Core history in one pipeline, since 2009	paywalls, logins or ads

WHY IT MATTERS

Bitcoin has no company behind it: no CEO, no board, no earnings call. Everything about it is public, from the code review behind each release to the channels of its Lightning payment network, but the record is scattered across GitHub, mailing lists, chat logs, network gossip and the chain itself.

So debates about software clients, centralization and funding run on opinion. We assemble the numbers once, with published methods, and keep them current for everyone who works on Bitcoin or reports on it.

- **Developers and maintainers:** review load, maintainer risk and who contributes to Bitcoin Core.
- **Node operators and implementers:** which software Lightning runs, and how fast it upgrades.
- **Funders:** where development effort and grant money go.
- **Journalists and researchers:** citable numbers with the source, the method and the code.

NEW RESEARCH · LIGHTNING SECURITY

PAPER 01 · WHO FAILS? — arXiv:2610.20952

Does Client Diversity Protect the Lightning Network?

The damage from a software failure depends on whether it reaches the network's hubs, not on how many nodes run the failing software.

- **84%** of public channel capacity runs on one implementation, LND
- **22%** of Core Lightning nodes went dark overnight, ten times the usual rate
- **4×** more reachability lost if ACINQ's Eclair node fails alone than if all 619 routable CLN nodes do

PAPER 02 · WHO PATCHES? — arXiv:2610.30350

Who Patches the Lightning Network?

Lightning operators upgrade when their platforms ship a release, not when a security advisory is published.

- **88 days** for half of LND nodes on an older line to upgrade after a release
- **1 of 12** security disclosures we could test measurably sped up upgrades
- **71%** of LND nodes exposed by Lightning Labs' August 2026 disclosures still ran an affected release two months later

WHAT WE RUN, EVERY DAY

PROGRAM 01

Bitcoin Core, measured

Who builds Bitcoin Core, who funds it, and whether it is healthy: a tracker, a monthly report, a weekly digest and a map of 7,790 contributors.

144 PRs merged in September 2026, +69% YoY

PROGRAM 02

Lightning & layer 2s

Every public Lightning node ranked and profiled daily since May 2023. L2 Watch audits sidechains, starting with Liquid.

12,536 nodes, **3,726 BTC** public capacity

PROGRAM 03

Research & forensics

Papers, and investigations such as the 1,983 BTC swept from weak-RNG addresses in 22 hours.

2 papers · **160+** mining pools profiled

PROGRAM 04

Learn

Open books and tools that take a newcomer from first principles to decoding real transactions.

3 open books · **5** interactive tools

DATA YOU CAN USE

DATASET	WHAT'S IN IT	UPDATED
`orange-dev-data`	Bitcoin Core commits, pull requests, reviews, BIPs, mailing-list and Delving Bitcoin threads, dev meetings, contributor identities	Weekly, since 2009
`lightning-data`	Lightning node ranks and profiles, channels, features, graph exports and weekly snapshots	Daily
`orange-eco-data`	955 Bitcoin open-source repositories and the 55,000+ developers behind them	Periodic
`bdl-report-assets`	Every published report slide and PDF, at a stable public URL	Every report
`Paper repositories`	The aggregate data and code behind every figure and number in our papers	Each version

Free to reuse
Datasets and report charts are CC BY 4.0: use them, including commercially, with credit.

Credit line
"Source: Bitcoin Data Labs" with a link to bitcoindatalabs.org or the dataset.

Custom cuts
Need a number we don't publish? Ask, and we'll send it with the method behind it.

STORIES THE DATA CAN SUPPORT

1. **Is Bitcoin Core development speeding up?**
   Monthly merges, review health and new contributors.
2. **Who pays Bitcoin's developers, and for what?**
   Funders, grantees and the projects they support.
3. **Does Lightning depend on one piece of software?**
   Paper 01: what the 2026 Core Lightning shutdown showed.
4. **Do node operators patch after an advisory?**
   Paper 02: the release line of every public node, 2023–2026.
5. **How concentrated is Bitcoin mining?**
   Every block since genesis attributed to its pool.
6. **What happens when wallet randomness is weak?**
   1,983 BTC swept from 4,925 addresses in 22 hours.

HOW WE WORK

Open by default
Open-source code, public data, no paywall or login.

Reproducible
Every number traces to a public pipeline and a documented method.

Honest about gaps
Missing data shows as "unavailable", never as a guess.

Responsible
Security results are published as totals; we never list vulnerable nodes.

Independent
No token, no sides. We report what the record shows.

WHERE WE'RE GOING

Our aim is a public annual report for Bitcoin: the same measured, reproducible record for every major implementation, every second layer and the chain itself, kept current and free to use.

- **Research.** Next in the series: what a force close costs, and how channels migrate if legacy signatures retire.
- **Coverage.** More implementations and Lightning clients, and more layers after Liquid.
- **Reach.** Every paper paired with a plain-language write-up for the people who run the software.

WHO WE ARE

Saurabh Kumar, founder. 15+ years leading data, analytics and experimentation: product optimization at PayPal, Head of Experimentation at realtor.com (News Corp), credit risk at HSBC. In Bitcoin since 2017. Author of the papers above and two open books.

A 501(c)(3) non-profit in Austin, Texas. Community partner: PlebLab. Code and data: github.com/bitcoindatalabs.

INTERVIEWS, DATA AND PARTNERSHIPS

Saurabh Kumar
sorukumar@bitcoindatalabs.org
x.com/sorukumar · linkedin.com/in/sorukumar

bitcoindatalabs.org · @Bitcoindatalabs · research